Jeremy D. Forsythe

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SUMMARY

Ecologist and data scientist specializing in remote sensing, wildfire, plant community assembly, ecosystem response to disturbance, & biogeochemical cycling. I integrate field measurements, eddy covariance, satellite data, & machine learning to understand functioning across terestrial ecosystems. Experienced in interdisciplinary collaboration, inclusive teaching, & curriculum development.

EDUCATION

Clemson University (Ph.D. Forestry & Environmental Conservation — May 2026) Clemson, SC

- **Award**: Phi Kappa Phi Honor Society
- **Coursework**: Data Science & Machine Learning, Statistical Methods, Quantitative Ecology, Micrometeorology, Scientific Writing

University of Kansas (M.A. Ecology & Evolutionary Biology — Aug 2018) Lawrence, KS

- **Award**: Department of Ecology & Evolutionary Biology Summer Fellowship
- **Coursework**: Biometry, Forest Ecosystems, Simulation Modeling, Plant Ecology, Likelihood Methods, Plant Systematics, Responsible Research & Teaching Effectiveness

University at Albany (B.S. Biological Sciences — Dec 2013) Albany, NY

- **Award**: Dean's List
- **Coursework**: Theoretical Ecology, Experimental Ecology, Biological Consequences of Climate Change, Geographic Information Systems, Statistical Inference, Evolution, Behavioral Ecology, Cell Biology, Genetics, Grazing in Terrestrial Ecosystems (Grad Class), Integrated Principles of Ecology (Grad Class), Field Ecology, Zoology, Animal Behavior, Organic Chemistry, Introductory Science Courses, Multi-Variate Calculus, Differential Equations

SKILLS

Technical: R, Python (scikit-learn, geopandas), Bash, GDAL, Javascript, Google Earth Engine, LaTeX, GIS (QGIS, ArcGIS), EddyPro, REddyProc

Remote Sensing & Data: Landsat, ERA5, ECOSTRESS, AppEEARS, Sentinel-2, MODIS, Flux Processing (QA/QC, Partitioning, Gap-filling), Vegetation Indices

Field & Instrumentation: Eddy Covariance Systems (LI-COR, Campbell, IRGASON), Environmental Sensor Installation/Maintenance (data loggers, instruments), FIA Vegetation Surveys, Identification of 200+ vascular and non-vascular species across Boreal, Midwest, and Southeastern U.S. ecoregions.

Professional: Scientific Writing, Public Speaking, Teaching, Mentoring, Interdisciplinary Collaboration

RESEARCH EXPERIENCE

Research Associate, NAU, NASA ABoVE, & Bonanza Creek LTER 2024–Present

- Liaison between Bonanza Creek LTER and NASA ABoVE, connecting field ecology and remote sensing teams.
- Quantified boreal reburn extent and analyzed fire–climate–vegetation interactions using Landsat, ERA5, and historical fire perimeters (1940–2023).
- Found that ∼20% of burned area since 1985 reburned within 70 years; showed high vapor pressure deficit amplifies fire size and severity along a continuum of time since burn.
- Contributed to meta-analysis evaluating satellite-based aboveground biomass estimates across Arctic and boreal regions.

Clemson University Ph.D. 2019–2026

- Dissertation
  - Examined drivers of photosynthetic carbon uptake, including diffuse radiation effects on canopy light distribution.
  - Found that diffuse radiation was the most influtential driver of daily light use efficiency in Southern pine forests using an intpretable machine learning framework (shapley).
  - Developed satellite derived flux upscaling tools combining tower data and vegetation indices using an optimized two-leaf LUE model.
  - Quantified the impact of light quality on the coupling of evapotranspiration and carbon fluxes, including the novel finding of diffuse fertilization on sapflow.
- Labwork
  - Built and maintained a 4-tower eddy covariance mesonet in coastal South Carolina (2019–2025), including instrumentation, calibration, and end-to-end data pipelines from raw measurements to QA/QC, gap-filled open access Ameriflux products, including off-grid solar power engineering, sensor telemetry, and field maintenance in remote coastal environments.

University of Kansas NSF REU & Master's Degree 2016–2018

- Designed independent research on native tallgrass prairie restoration across increasing spatial scales.
- Studied long-term forest succession (woody invasion) across a prairie–forest ecotone.
- Applied maximum likelihood methods to model forest stand stability and predict future vegetation change.

University at Albany

- Developed a biological control invasive species management project using ungulate grazing.

TEACHING & OUTREACH

Northern Arizona University

- Instructor of Record: *Observing Earth from Above* satellite remote sensing independent study.
- Guest Lecturer: Machine learning and regression methods in ecological data analysis courses (Dr. Xanthe Walker).
- Organizer: Machine Learning for Ecologists workshop at the 2026 Ecological Society of America meeting.
- Founded "Snack Overflow," an inclusive coding and writing working group for undergraduates and graduate students.
- Presented research biannually at the Alaska Fire Science Consortium.
- Represented NASA ABoVE and NSF at the Interagency Arctic Research Policy Committee (IARPC) meeting connecting tribal leaders with researchers.

Independent Contractor, Chapman University & NASA ECOSTRESS 2022–2024

- Co-developed and instructed the ICE CREAM (Integrating Communication of ECOSTRESS into Community Research, Education, Applications, and Media) open-access satellite remote sensing curriculum with Drs. Joshua Fisher and Gregory Goldsmith; adopted by 12+ minority-serving institutions and accessed by 1,400+ users.
- Created tutorials using NASA AppEEARS and QGIS; delivered lectures and mentored students.
- Independent assessment showed significant gains in science interest (20%), belonging (11%), science identity (9%), and content understanding (19%) across diverse student populations.

Clemson University

- Mentored undergraduate researchers through Clemson's UPIC Program, advising 1–2 students each summer.

- Guest Lecturer: Data management and visualization in R; invasive plant species ecology in wetland biology.
- Founded "Snack Overflow" coding and writing working group (later implemented at NAU).
- Active member of Ameriflux DEI & FLUXNET Education Committees.

University of Kansas

- Teaching Assistant (2 semesters): BIO 152 Principles of Biology. Collaborated on flipped classroom redesign with Dr. Mark Mort; authored exams, lectured, and ran lab sections.
- Contributed to course redesign emphasizing active learning strategies including learning groups, real-time response surveys, and integrated online tools.
- Mentored NSF REU cohort of 12 students through a 10-week independent research program.

SERVICE

Peer Reviewer: *Arctic, Antarctic, and Alpine Research*; *International Journal of Climatology*; *Ecosystems* (Springer)

Committee Membership:

- Ameriflux DEI Committee
- FLUXNET Education Committee

REFERENCES

- Dr. Michelle Mack: Research Associate Advisor; Regents Professor at Northern Arizona University.
  michelle.mack@nau.edu
- Dr. Xanthe Walker: Research Associate Advisor; Assistant Professor at Northern Arizona University.
  xanthe.walker@nau.edu
- Dr. Scott Goetz: Research Associate Advisor; Regents Professor at Northern Arizona University.
  scott.goetz@nau.edu
- Dr. Joshua Fisher: Mentor; Presidential Fellow of Ecosystem Science at Chapman University.
  jbfisher@chapman.edu
- Dr. Gregory Goldsmith: Mentor; Professor at Chapman University.
  goldsmit@chapman.edu
- Dr. Tom O'Halloran: Ph.D. Advisor; Associate Professor at Clemson University.
  tohallo@clemson.edu
- Dr. Bryan Foster: Master's Advisor, REU Mentor, and University of Kansas Faculty.
  bfoster@ku.edu

PUBLICATIONS

- {Under review at *Fire Ecology*} Forsythe, J.D., Mack, M., Berner, L., Goetz, S., Hansen, W., Hayes, K., Johnstone, J., Potter, S., Rogers, B., Talucci, A., Wang, J., & Walker, X. Short-Interval Wildfires Are Smaller, Less Severe, and Require More Extreme Weather than Long-Interval Fires in Boreal North America.
- {Under review at *JGR Biogeosciences*} Forsythe, J.D. & O'Halloran, T.L. Light quality controls daily LUE in Southern pine forests: an interpretable machine learning approach.
- {In Preparation} Forsythe, J.D., O'Halloran, T.L., & Hawman, P. Satellite data infused light use efficiency models reveal importance of diffuse light fertilization in U.S. Southern pine ecosystems. *Agricultural and Forest Meteorology.*
- {In Preparation} Forsythe, J.D., O'Halloran, T.L., Thomas, R.Q., Fisher, J.B., & Duberstein, J. Diffuse light increases evapotranspiration and water use efficiency in Southern pine. *Ecohydrology.*
- Liang, W., Bauck, K., Maloney, D., Hoy, E., Montesano, P., Lara, M.J., Yang, D., Forsythe, J., Duncanson, L., Walker, X., & Wang, J.A. (2026). Meta-analysis of North American Arctic and boreal aboveground biomass datasets: assessing accuracy, dynamics, and similarities. *Environmental Research Letters*, 21, 054004.

- Zhu, Q., Chen, J., Bourque, C.P.A., et al. including Forsythe, J.D. (2024). Albedo-induced global warming potential following disturbances in global temperate and boreal forests. *JGR Biogeosciences*, 129.
- Aguilos, M., Sun, G., Liu, N., et al. including Forsythe, J. (2024). Energy availability and leaf area dominate control of ecosystem evapotranspiration in the southeastern U.S. *Agricultural and Forest Meteorology*, 349.
- Williams, T.M., B. Williams, B. Song, T.L. O'Halloran, J.D. Forsythe. 2022. Mapping natural forest stands with low-cost drones. *Math. and Comp. Forestry & Nat.-Res. Sci.*, 14(1), 22–42.
- Ahlswede, B.J., O'Halloran, T.L., Forsythe, J.D., & Thomas, R.Q. 2021. A minimally managed switchgrass ecosystem in a humid subtropical climate is a source of carbon to the atmosphere. *GCB Bioenergy*, 14, 24–36.
- Williams, T.M., T.L. O'Halloran, B. Song, J.D. Forsythe, and B.J. Williams. 2021. Sources of error from dense understory vegetation in Coastal Plain forest hydrologic analyses using LiDAR DEMs. *In Proc. 13th Southern Forestry and Natural Resource Management GIS Conf.* Athens, GA.
- Williams, T.M., T.L. O'Halloran, B. Song, J.D. Forsythe, and B.J. Williams. 2020. Evaluating high water table hydrology and eddy covariance measurements of evapotranspiration at a newly instrumented watershed in coastal South Carolina. *In Proc. 7th Interagency Conf. on Research in the Watersheds.* Tifton, GA.
- Forsythe, J.D., O'Halloran, T.L., & Kline, M.A. 2020. An eddy covariance mesonet for measuring greenhouse gas fluxes in coastal South Carolina. *Data*, 5, 1–20.

PRESENTATIONS

- **Poster:** Goldsmith GR, Verbeke M, Forsythe J, Fisher J. A distributed model for undergraduate education in environmental remote sensing: increased student interest in science and sense of science identity and belonging. European Geosciences Union (EGU) General Assembly; 05/2026.
- **Talk:** Forsythe J, Potter S, Rogers B, Berner L, Goetz S, Johnstone J, Mack M, Walker X. Quantifying The Controls On Boreal Forest Reburning With Interpretable Machine Learning. American Geophysical Union (AGU); 2025.
- **Talk:** Hall J, Hayes K, Asena Q, Forsythe J, et al. Predicting forest burn risk from 2026 to 2040 in interior Alaska. American Geophysical Union (AGU); 2025.
- **Talk:** Liang W, Bauck K, Maloney D, et al. including Forsythe J. Meta-analysis of North American Arctic and Boreal Aboveground Carbon Maps to Provide Guidance for User Communities. American Geophysical Union (AGU); 2025.
- **Talk:** Forsythe J, Potter S, Rogers B, Berner L, Goetz S, Johnstone J, Mack M, Walker X. Legacy of Wildfire in Boreal Forests: Characterizing Fire-on-Fire Interactions with Satellite Remote Sensing. Bonanza Creek LTER All Hands Meeting; 06/2025.
- **Talk:** Forsythe J, Rogers B, Berner L, Goetz S, Johnstone J, Mack M, Walker X. Drivers and Impacts of Reburning in Boreal Forest Ecosystems (DIRE). NASA ABoVE Team Meeting; 05/2025.
- **Poster:** Maloney D, Hoy E, Larson E, Montesano P, Wang J, Liang W, Bauck K, Walker X, Forsythe J. A Comparative Analysis of Available Arctic and Boreal Biomass Datasets. NASA ABoVE Team Meeting; 05/2025.
- **Invited Talk:** Forsythe J, Mack M, Walker X. Decades of Discovery: How NSF's Long Term Ecological Research is Advancing Arctic Understanding. Interagency Arctic Research Policy Committee (IARPC); 2025.
- **Invited Talk:** Forsythe J, Potter S, Rogers B, Berner L, Goetz S, Johnstone J, Mack M, Walker X. Mapping Top-Down and Bottom-Up Controls on Wildfire in Boreal Forest Ecosystems. Interagency Arctic Research Policy Committee (IARPC); 2025.
- **Talk:** Forsythe J, Potter S, Rogers B, Berner L, Goetz S, Johnstone J, Mack M, Walker X. Top-Down and Bottom-Up Controls on Short Interval Wildfire in Boreal Forest Ecosystems. American Geophysical Union (AGU); 2024.

- **Invited Talk:** Forsythe J. Pixels, Patterns, and Plants: Integrating Remote Sensing, Micrometeorology, and Field Ecology to Unravel Ecosystem Functioning Across Diverse Landscapes. NAU Ecoinformatics Seminar Series; 2024.
- **Poster:** Forsythe J, Potter S, Rogers B, Berner L, Goetz S, Johnstone J, Mack M, Walker X. Evidence of Reburning in Boreal Forest Ecosystems from Landsat Observations. NASA ABoVE Team Meeting; 05/2024.
- **Invited Talk:** Forsythe JD, O'Halloran TL. Diffuse Light & Evapotranspiration in Southern Pine Forests. NASA ECOSTRESS Science and Application Team Meeting; Fall 2023.
- **Invited Talk:** Forsythe JD. Writing Tutorials For Accessing & Visualizing ECOSTRESS Data With Open Source Tools. NASA ECOSTRESS Science and Application Team Meeting; Fall 2023.
- **Talk:** Forsythe JD, O'Halloran TL. Responses of Southern Pine Forest Gross Primary Productivity to Diffuse Radiation: A 17 Site-Year Study. Ecological Society of America; 08/2023.
- **Talk:** Forsythe JD, O'Halloran TL. Sensitivity of Diffuse Light Fertilization in Southern Pine Carbon Fluxes Observed with Remotely-Sensed Vegetation Indices. American Meteorological Society 35th Conference; 05/2023.
- **Poster:** Forsythe JD, O'Halloran TL, Wise M, Balkcum O, Song B, Kline MA, DeGarady C. How do fire & management alter longleaf pine understory carbon stocks & whole ecosystem carbon sequestration? 14th Biennial Longleaf Conference; 10/2022.
- **Poster:** Forsythe JD, O'Halloran TL, Williams T, Kaminski R, Kline MA. An Eddy Covariance Mesonet Measuring Coastal Carbon Fluxes in South Carolina. 7th North American Carbon Program OSM; 03/2021.
- **Poster:** Forsythe JD, O'Halloran TL, Kline MA. Establishing An Eddy Covariance Mesonet in Coastal South Carolina. Ameriflux; 2020.
- **Poster:** Forsythe JD, Foster BL. The Effects Of Disturbance And Soil Nutrient Enrichment On Grassland Community Biodiversity Across Spatial Scales. KU Madison & Lila Self Graduate Fellowship Symposium.